Mathematical Equations for the Parametric Bird

A Complete Formulation of the Wireframe Bird Image

Generated by: `bird.py` — Surface Model v7

Master Surface Equation

The entire bird is assembled from **tube-swept parametric surfaces**. Every component curve $\boldsymbol{\gamma}(t)$ generates a surface by sweeping a circle of radius $r(t)$ perpendicular to the curve:

$$\boxed{\boldsymbol{S}(t,\theta) \;=\; \boldsymbol{\gamma}(t) \;+\; r(t)\Big[\,\hat{\boldsymbol{N}}(t)\cos\theta + \hat{\boldsymbol{B}}(t)\sin\theta\,\Big]} \qquad t\in[0,1], \quad \theta\in[0,2\pi] \tag{1}$$

where $\hat{\boldsymbol{N}}(t)$ and $\hat{\boldsymbol{B}}(t)$ are the *normal* and *binormal* vectors of the **Rotation-Minimising Frame** (RMF) described in Section 3.

Bézier Curve Primitives

All spine curves $\boldsymbol{\gamma}(t)$ are constructed from Bézier curves over control points $\boldsymbol{P}_i \in \mathbb{R}^3$.

Quadratic Bézier Curve

$$\boldsymbol{B}_2(t;\,\boldsymbol{P}_0,\boldsymbol{P}_1,\boldsymbol{P}_2) \;=\; (1-t)^2\,\boldsymbol{P}_0 + 2(1-t)t\,\boldsymbol{P}_1 + t^2\,\boldsymbol{P}_2, \qquad t\in[0,1] \tag{2}$$

Cubic Bézier Curve

$$\boldsymbol{B}_3(t;\,\boldsymbol{P}_0,\boldsymbol{P}_1,\boldsymbol{P}_2,\boldsymbol{P}_3) \;=\; (1-t)^3\,\boldsymbol{P}_0 + 3(1-t)^2t\,\boldsymbol{P}_1 + 3(1-t)t^2\,\boldsymbol{P}_2 + t^3\,\boldsymbol{P}_3, \qquad t\in[0,1] \tag{3}$$

Rotation-Minimising Frame (RMF)

The standard Frenet–Serret frame is undefined where curvature vanishes. We use the **Rotation-Minimising Frame** (Bishop frame), propagated iteratively:

Step 1 — Unit Tangent.

$$\hat{\boldsymbol{T}}(t_i) \;=\; \frac{\left.\dfrac{d\boldsymbol{\gamma}}{dt}\right|_{t_i}}{\left\|\left.\dfrac{d\boldsymbol{\gamma}}{dt}\right|_{t_i}\right\|} \tag{4}$$

Computed numerically via central differences $\left.\dfrac{d\boldsymbol{\gamma}}{dt}\right|_{t_i} \approx \dfrac{\boldsymbol{\gamma}(t_{i+1}) - \boldsymbol{\gamma}(t_{i-1})}{t_{i+1} - t_{i-1}}$.

Step 2 — Initial Normal. Choose an arbitrary reference $\boldsymbol{e}$ (e.g. $\boldsymbol{e} = (0, 1, 0)$) not parallel to $\hat{\boldsymbol{T}}(t_0)$:

$$\hat{\boldsymbol{N}}(t_0) = \frac{\hat{\boldsymbol{T}}(t_0) \times \boldsymbol{e}}{\|\hat{\boldsymbol{T}}(t_0) \times \boldsymbol{e}\|}, \qquad \hat{\boldsymbol{B}}(t_0) = \hat{\boldsymbol{T}}(t_0) \times \hat{\boldsymbol{N}}(t_0) \tag{5}$$

Step 3 — Parallel Transport (RMF propagation). For $i = 1, 2, \ldots, n-1$:

$$\boldsymbol{N}_i^{\perp} = \hat{\boldsymbol{N}}(t_{i-1}) - \left[\hat{\boldsymbol{N}}(t_{i-1}) \cdot \hat{\boldsymbol{T}}(t_i)\right] \hat{\boldsymbol{T}}(t_i) \tag{6}$$

$$\hat{\boldsymbol{N}}(t_i) = \frac{\boldsymbol{N}_i^{\perp}}{\|\boldsymbol{N}_i^{\perp}\|} \tag{7}$$

$$\hat{\boldsymbol{B}}(t_i) = \frac{\hat{\boldsymbol{T}}(t_i) \times \hat{\boldsymbol{N}}(t_i)}{\|\hat{\boldsymbol{T}}(t_i) \times \hat{\boldsymbol{N}}(t_i)\|} \tag{8}$$

Wing Surface Equations

Each wing is modelled as a **bilinear ruled surface** stretched between a *leading-edge* curve $\boldsymbol{\alpha}(u)$ and a *trailing-edge* curve $\boldsymbol{\beta}(u)$, both cubic Bézier curves in $\mathbb{R}^3$.

Ruled Wing Surface

$$\boxed{\boldsymbol{S}_{\text{wing}}(u, v) = (1 - v)\,\boldsymbol{\alpha}(u) + v\,\boldsymbol{\beta}(u), \qquad u, v \in [0, 1]} \tag{9}$$

$$\begin{array}{ll} u = 0 & \text{wing root (body junction)} \\ u = 1 & \text{wing tip} \\ v = 0 & \text{leading edge} \\ v = 1 & \text{trailing edge} \end{array}$$

Left Wing Leading Edge

$$\boldsymbol{\alpha}_L(u) = \boldsymbol{B}_3\left(u;\ \begin{pmatrix} 0.00 \\ 0.10 \\ 0.10 \end{pmatrix}, \begin{pmatrix} -1.80 \\ 0.05 \\ 0.55 \end{pmatrix}, \begin{pmatrix} -3.80 \\ -0.08 \\ 0.60 \end{pmatrix}, \begin{pmatrix} -5.50 \\ -0.20 \\ 0.15 \end{pmatrix}\right) \tag{10}$$

Left Wing Trailing Edge

$$\boldsymbol{\beta}_L(u) = \boldsymbol{B}_3\left(u;\ \begin{pmatrix} 0.00 \\ 0.10 \\ -0.22 \end{pmatrix}, \begin{pmatrix} -1.40 \\ 0.00 \\ -0.05 \end{pmatrix}, \begin{pmatrix} -3.60 \\ -0.12 \\ 0.25 \end{pmatrix}, \begin{pmatrix} -5.40 \\ -0.20 \\ 0.12 \end{pmatrix}\right) \tag{11}$$

Right Wing (Bilateral Symmetry)

The right wing is the reflection of the left wing across the sagittal plane ($x = 0$):

$$\boldsymbol{\alpha}_R(u) = \boldsymbol{M}\,\boldsymbol{\alpha}_L(u), \qquad \boldsymbol{\beta}_R(u) = \boldsymbol{M}\,\boldsymbol{\beta}_L(u), \qquad \boldsymbol{M} = \operatorname{diag}(-1,\ 1,\ 1) \tag{12}$$

Wing Mesh Lines

The wing surface is rendered as two families of iso-parameter lines:

$$\begin{aligned}
\text{Span lines (chord-dir):} \quad \mathcal{L}_u &= \{\boldsymbol{S}(u_0, v) \;:\; v \in [0,1]\} \quad \text{for fixed } u_0 && (13) \\
\text{Chord lines (span-dir):} \quad \mathcal{L}_v &= \{\boldsymbol{S}(u, v_0) \;:\; u \in [0,1]\} \quad \text{for fixed } v_0 && (14)
\end{aligned}$$

Body / Torso

The torso is a **swept tube** along a cubic Bézier spine from head to tail:

Body Spine Curve

$$\boldsymbol{\gamma}_{\text{body}}(t) = \boldsymbol{B}_3\left(t;\ \begin{pmatrix}0\\0\\0.90\end{pmatrix}, \begin{pmatrix}0\\0.05\\0.50\end{pmatrix}, \begin{pmatrix}0\\0\\-0.15\end{pmatrix}, \begin{pmatrix}0\\-0.05\\-0.70\end{pmatrix}\right) \tag{15}$$

Body Radius Profile

The radius follows an **asymmetric sine-power** profile, thicker at the breast and tapering at head and tail:

$$r_{\text{body}}(t) = R_0 + A \sin^{\alpha}(\pi t), \qquad t \in [0,1] \tag{16}$$

Parameter	Symbol	Value
Base radius	R_0	0.06
Amplitude	A	0.28
Shape exponent	α	1.2

The body surface is then:

$$\boldsymbol{S}_{\text{body}}(t, \theta) = \boldsymbol{\gamma}_{\text{body}}(t) + r_{\text{body}}(t)\left[\hat{\boldsymbol{N}}(t)\cos\theta + \hat{\boldsymbol{B}}(t)\sin\theta\right] \tag{17}$$

Head Spike

The beak/head is a **tapered cone** swept along a straight line above the body:

Spike Axis

$$\boldsymbol{\gamma}_{\text{spike}}(t) = (1-t)\,\boldsymbol{p}_{\text{head}} + t\,\boldsymbol{p}_{\text{tip}}, \qquad t \in [0,1] \tag{18}$$

where $\boldsymbol{p}_{\text{head}} = \boldsymbol{\gamma}_{\text{body}}(0)$ and $\boldsymbol{p}_{\text{tip}} = \boldsymbol{p}_{\text{head}} + (0,\ 0,\ 0.45)^{\top}$.

Spike Radius Profile

The radius tapers to zero at the tip via a power law:

$$r_{\text{spike}}(t) = R_{\text{spike}}\,(1-t)^{p}, \qquad R_{\text{spike}} = 0.10, \quad p = 1.4 \tag{19}$$

Tail Feathers

M tail feathers fan out below the body in a symmetric spread. Each feather $j = 0, \dots, M-1$ is a swept tube along a **quadratic Bézier curve**.

Azimuth Angle

$$\phi_j = \phi_{\min} + \frac{j}{M-1}\,(\phi_{\max} - \phi_{\min}), \qquad \phi_{\min} = 200^\circ, \quad \phi_{\max} = 340^\circ \tag{20}$$

Feather Length (Bell Curve)

$$L_j = L_{\text{side}} + \big(L_{\text{mid}} - L_{\text{side}}\big)\sin\!\left(\pi\,\frac{j}{M-1}\right), \qquad L_{\text{mid}} = 2.2, \quad L_{\text{side}} = 0.80 \tag{21}$$

Feather Spine (Quadratic Bézier)

$$\boldsymbol{p}_{\text{tip},j} = \boldsymbol{p}_{\text{tail}} + L_j \begin{pmatrix} \cos\phi_j \\ 0 \\ \sin\phi_j \end{pmatrix} \tag{22}$$

$$\boldsymbol{p}_{\text{ctrl},j} = \tfrac{1}{2}(\boldsymbol{p}_{\text{tail}} + \boldsymbol{p}_{\text{tip},j}) + \begin{pmatrix} 0.10\,L_j\cos(\phi_j + \frac{\pi}{2}) \\ 0.04\,L_j \\ 0.10\,L_j\sin(\phi_j + \frac{\pi}{2}) \end{pmatrix} \tag{23}$$

$$\boldsymbol{\gamma}_j(t) = \boldsymbol{B}_2(t;\ \boldsymbol{p}_{\text{tail}},\ \boldsymbol{p}_{\text{ctrl},j},\ \boldsymbol{p}_{\text{tip},j}) \tag{24}$$

Feather Radius Profile

$$r_j(t) \;=\; R_{0,T}\,(1-t)^{\beta_T} + R_{\min,T}, \qquad R_{0,T} = 0.18, \quad \beta_T = 0.55, \quad R_{\min,T} = 0.008 \tag{25}$$

Summary of All Equations

Component	Governing Equation
Master surface	$\boldsymbol{S}(t,\theta) = \boldsymbol{\gamma}(t) + r(t)\big[\hat{\boldsymbol{N}}\cos\theta + \hat{\boldsymbol{B}}\sin\theta\big]$
Quadratic Bézier	$\boldsymbol{B}_2(t) = (1-t)^2\boldsymbol{P}_0 + 2(1-t)t\,\boldsymbol{P}_1 + t^2\boldsymbol{P}_2$
Cubic Bézier	$\boldsymbol{B}_3(t) = (1-t)^3\boldsymbol{P}_0 + 3(1-t)^2t\,\boldsymbol{P}_1 + 3(1-t)t^2\boldsymbol{P}_2 + t^3\boldsymbol{P}_3$
RMF transport	$\hat{\boldsymbol{N}}_i = \dfrac{\hat{\boldsymbol{N}}_{i-1} - (\hat{\boldsymbol{N}}_{i-1}\cdot\hat{\boldsymbol{T}}_i)\hat{\boldsymbol{T}}_i}{\lVert\cdots\rVert},\quad \hat{\boldsymbol{B}}_i = \hat{\boldsymbol{T}}_i \times \hat{\boldsymbol{N}}_i$
Wing surface	$\boldsymbol{S}_{\text{wing}}(u,v) = (1-v)\boldsymbol{\alpha}(u) + v\,\boldsymbol{\beta}(u)$
Body radius	$r_{\text{body}}(t) = R_0 + A\sin^{\alpha}(\pi t)$
Spike radius	$r_{\text{spike}}(t) = R_{\text{spike}}\,(1-t)^{1.4}$
Tail azimuth	$\phi_j = \phi_{\min} + \frac{j}{M-1}(\phi_{\max} - \phi_{\min})$
Tail length	$L_j = L_{\text{side}} + (L_{\text{mid}} - L_{\text{side}})\sin\!\left(\frac{\pi j}{M-1}\right)$
Tail radius	$r_j(t) = R_{0,T}(1-t)^{\beta_T} + R_{\min,T}$
Right-wing sym.	$\boldsymbol{\alpha}_R(u) = \mathrm{diag}(-1,1,1)\,\boldsymbol{\alpha}_L(u)$

Rendering Parameters

Parameter	Symbol	Value
Span-wise sample lines	N_{span}	110
Chord-wise sample lines	N_{chord}	75
Points per line	N_{pts}	200
Body ring count	N_{body}	140
Rings per cross-section	N_{θ}	28–38
Tail feather count	M	18
Tail sample count	N_{tail}	120
Line width (wings)	ℓ_w	0.25 pt
Camera elevation	ϵ	18°
Camera azimuth	ψ	−90°

All coordinates are in arbitrary units; the bird spans approximately $[-5.5, 5.5]$ *in the* x*-axis (wing span).*